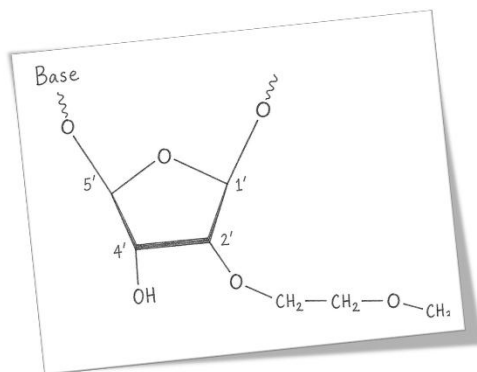
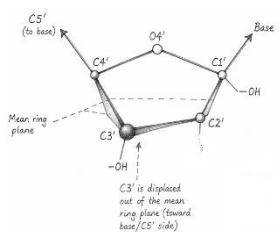


2'-O-Methoxyethyl (2'-MOE) / MOE chemistry



In 1995, Pierre Martin, a chemist at Ciba-Geigy in Basel, Switzerland, reported the synthesis and properties of 2'-O-alkylated ribonucleosides, including the 2'-O-(2-methoxyethyl) modification, in *Liberation Serif Chimica Acta* (Martin, 1995; Hill & Hall, 2023). Martin's work was motivated by the need for ribose modifications that were compatible with solid-phase oligonucleotide synthesis and that could protect oligonucleotides from nuclease degradation without compromising hybridization to RNA. His synthetic approach provided access to 2'-O-alkylribonucleosides in which the ribose 2'-hydroxyl group was replaced with ethylene glycol-derived substituents, including what is now known as the MOE group. Oligonucleotides containing this modification displayed what Martin described as “surprising properties”: increased affinity and specificity for RNA targets, as well as enhanced resistance to nuclease digestion.



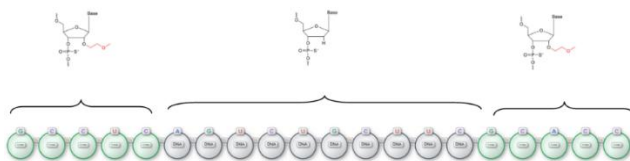
The beneficial properties of MOE-modified oligonucleotides are closely linked to their effects on duplex structure and stability. Martin's original studies showed that incorporation of MOE residues increased melting temperature against complementary RNA by approximately 0.9–1.6 °C per modification, while also improving discrimination against

mismatched RNA sequences. Subsequent structural studies helped explain these observations by showing that the MOE substituent favors a C3'-endo ribose sugar pucker, a conformation associated with A-form RNA duplexes. This preorganization enhances RNA binding while maintaining sequence specificity. Importantly, MOE modification also improves nuclease resistance; in serum stability experiments, MOE-containing oligonucleotides persisted far longer than unmodified or 2'-O-methyl-modified comparators. Together, these features made MOE chemistry particularly attractive for therapeutic ASO design.

As antisense technology evolved, MOE chemistry became a defining feature ASOs, especially when combined with phosphorothioate (PS) backbone modification. Early PS ASOs improved nuclease stability and tissue distribution but were limited by reduced RNA-binding affinity and pro-inflammatory liabilities. Incorporation of MOE-modified nucleotides helped address these shortcomings by increasing potency, extending tissue residence time, and improving tolerability. In the commonly used “gapmer” architecture, MOE-modified nucleotides are placed in the flanking “wings” of an ASO to enhance affinity and stability, while a central DNA gap supports RNase H-mediated cleavage of the target RNA. MOE chemistry is also used in fully modified splice-switching oligonucleotides, where the goal is not RNA degradation but modulation of pre-mRNA splicing.

These combined chemical innovations enabled the clinical translation of MOE-modified ASOs and helped establish oligonucleotides as a major therapeutic modality. MOE chemistry is present in multiple approved drugs, including mipomersen, nusinersen, inotersen, and volanesorsen, and it remains a foundation for numerous candidates in clinical development. Mipomersen provided an early proof of concept for RNase H-recruiting MOE gapmers capable of reducing disease-associated protein expression through mRNA targeting, while nusinersen demonstrated the therapeutic power of fully modified MOE ASOs for splice correction in spinal muscular atrophy. More broadly, the success of MOE-modified ASOs illustrates how rational chemical modification can transform fragile nucleic acids into drug-like molecules with tunable activity, durability, and tissue distribution.

Inotersen



Inotersen is administered as a once-weekly subcutaneous injection and has demonstrated clinical efficacy in phase II/III NEURO-TTR randomized controlled trial. The study showed that inotersen significantly slowed the progression of neuropathy and improved quality of life compared with placebo, as measured by the modified Neuropathy Impairment Score and the Norfolk Quality of Life–Diabetic Neuropathy questionnaire (CADTH, 2020). However, despite achieving statistical significance, the clinical meaningfulness of these improvements remains uncertain due to the absence of clearly defined thresholds for clinically important differences (CADTH, 2020).

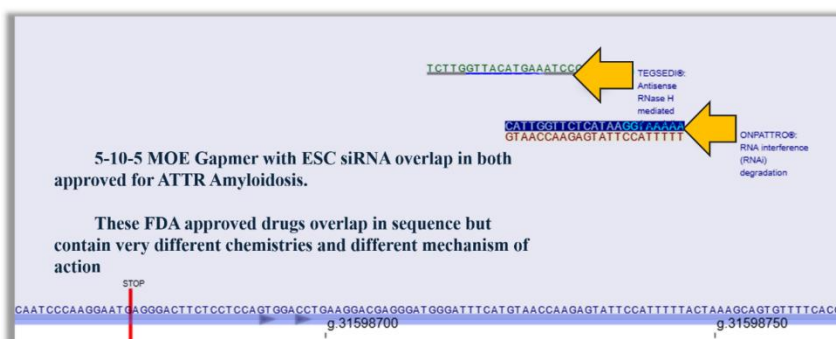
Safety remains a key concern, as treatment is associated with a high incidence of adverse events, including thrombocytopenia, renal impairment, and injection-site reactions. Serious adverse events and deaths have been reported in clinical trials, necessitating regular monitoring of platelet counts and renal function (CADTH, 2020). Clinical guidance further emphasizes the importance of ongoing monitoring through regular blood and urine tests to assess both efficacy and toxicity (Mayo Clinic, n.d.).

Inotersen can reduce platelet counts, increasing the risk of bleeding and infection, and patients are advised to take precautions to minimize injury and exposure to infectious agents (Mayo Clinic, n.d.). It has also been associated with serious organ-related complications, including kidney and liver toxicity, with symptoms such as reduced urine output, swelling, jaundice, or abdominal pain requiring immediate medical attention (Mayo Clinic, n.d.). In addition, there is an elevated risk of inflammatory or immune-mediated conditions, stroke, and arterial dissection, highlighting the need for vigilance regarding neurological symptoms (Mayo Clinic, n.d.).

Hypersensitivity reactions, including life-threatening allergic responses, have also been reported, alongside reductions in vitamin A levels that may lead to visual disturbances, necessitating ophthalmologic monitoring in some patients (Mayo Clinic, n.d.). Patients are further advised to consult their healthcare provider before taking any concomitant

medications, including over-the-counter or herbal products, due to potential interactions (Mayo Clinic, n.d.).

Overall, Inotersen represents an important advancement in RNA-targeting therapeutics and offers a more convenient, self-administered alternative to intravenous treatments. However, its use is limited by safety concerns and ongoing uncertainties regarding long-term efficacy and broader clinical benefits (CADTH, 2020; Rüger et al., 2020).



MOE-modified ASOs are widely used in two RNA-targeting strategies: RNase H-mediated RNA degradation and splice modulation via steric blocking. These two approaches enable treatment of a broad range of genetic diseases (Sang et al., 2024). For example, MOE-containing gapmer ASOs such as Inotersen and Eplontersen are used in the treatment of hereditary transthyretin amyloidosis (hATTR), where they bind transthyretin (TTR) mRNA and promote its degradation, thereby reducing pathogenic protein accumulation (Sang et al., 2024). Similarly, Tofersen, a 5-10-5 MOE-modified gapmer ASO approved for amyotrophic lateral sclerosis (ALS), targets mutant SOD1 mRNA, demonstrating the utility of MOE chemistry in neurodegenerative disease through selective reduction of toxic protein expression (Sang et al., 2024). In the context of splice modulation, Nusinersen (Spinraza) discovered at Cold Spring Harbor uses a fully MOE-modified ASO to promoting inclusion of exon 7 in SMN2 mRNA and thereby restoring production of functional SMN protein (Sang et al., 2024). Beyond these established therapies, MOE chemistry has also enabled highly personalized approaches, exemplified by Milasen, a patient-specific ASO designed to correct aberrant splicing in Batten disease, highlighting the adaptability of MOE-modified oligonucleotides for precision medicine (Sang et al., 2024). This chemistry is heavily used by nLorem for their nano-rare therapeutics programs.

Incorporation of MOE at the 2'-position of the ribose increases conformational rigidity and enhances hybridization while reducing susceptibility to nuclease-mediated cleavage (Zanardi et al., 2018). From a pharmacokinetic and pharmacodynamic (PK/PD) perspective, MOE-modified ASOs exhibit slow plasma clearance, high plasma protein binding, and extensive tissue distribution—particularly to the liver, kidney, and adipose tissue—resulting in long tissue half-lives that can extend from weeks to months (Takakusa et al., 2023). For example, 2'-MOE ASOs such as Mipomersen demonstrate terminal half-lives of 30 days in plasma and similar persistence in liver tissue, reflecting equilibrium between plasma and tissue compartments and supporting infrequent dosing regimens (Takakusa et al., 2023). These prolonged kinetics arise from strong protein binding and intracellular retention following endocytosis and endosomal trafficking, which enable sustained pharmacological activity even after plasma levels decline (Takakusa et al., 2023). Importantly, pharmacodynamic effects are more closely correlated with tissue concentrations than plasma levels, highlighting the importance of tissue PK in dose optimization and therapeutic efficacy (Christensen et al., 2025). Furthermore, advances in chemical stabilization and conjugation strategies have enabled improved delivery beyond the liver, expanding the therapeutic reach of oligonucleotides to extrahepatic tissues such as muscle, brain, and lung (Anand et al., 2025).

Fomivirsen (Vitraven[®])	Phosphorothioate backbone (PS)	Cytomegalovirus (CMV) retinitis	1999
Mipomersen (Kynamro[®])	PS & 2'-MOE	Homozygous familial hypercholesterolemia	2013
Nusinersen (Spinraza[®])	PS & 2'-MOE	Spinal muscular atrophy	2016
Inotersen (Tegsedi[®])	PS & 2'-MOE	Hereditary transthyretin-mediated amyloidosis	2018
Milasen	PS & 2'-MOE	CLN7 gene mutation associated with Batten disease	2018
Volanesorsen (Waylivra[®])	2'-MOE	Familial chylomicronemia syndrome	EMA approved 2019
Tofersen (Qalsody[®])	PS & 2'-MOE	Amyotrophic lateral sclerosis with SOD1 mutation	2023
Eplontersen (Wainua[®])	PS, 2'-MOE & GalNAc	Polyneuropathy of hereditary transthyretin-mediated amyloidosis	2023

Volanesorsen (Waylivra)

Volanesorsen (Waylivra): ApoC-III targeting ASO for familial chylomicronemia syndrome (FCS)

Base:	A G mC mU mU mC T T G T mC mC A G mC mU mU mU A mU
Backbone	Full phosphorothioate (PS)
Sugar:	E E E E E d d d d d d d d d d E E E E E

legend: E = 2'-MOE d = DNA (2'-deoxy)

Volanesorsen demonstrated substantial triglyceride-lowering efficacy but is associated with several safety considerations. The most clinically significant risk is thrombocytopenia, including cases of severe platelet reduction. In addition, injection-site reactions are very common, occurring in a substantial proportion of administrations, although these are generally mild to moderate and primarily affect tolerability rather than safety. Rare but serious immune-mediated events, such as serum sickness, have also been reported, indicating potential for hypersensitivity reactions. (Witztum et al., 2019; Tardif et al., 2016).

Olezarsen, a GalNAc-conjugated version of the Volanesorsen ASO was developed to solve its relatively poor safety profile. Thrombocytopenia with Volanesorsen was attributed to high systemic exposure and dosing burden. By incorporating N-acetylgalactosamine (GalNAc) for hepatocyte-targeted delivery, Olezarsen matches efficacy with less frequent doses, thereby reducing systemic exposure. Clinical data demonstrate that while some adverse events persist, including mild thrombocytopenia and liver enzyme elevations, these effects are dose-dependent and more pronounced at higher doses, with overall safety outcomes approaching those of placebo. This supports the concept that dose and exposure are central determinants of antisense oligonucleotide toxicity, and that GalNAc conjugation can effectively improve the therapeutic index by decoupling efficacy from systemic toxicity (Marston et al., 2026; Witztum et al., 2019).

	Volanesorsen	Olezarsen (GalNAc)
Dosing	High dose (e.g., 300 mg weekly → biweekly)	Lower dose (50–80 mg monthly)
Delivery	Systemic (no targeting)	Hepatocyte-targeted (GalNAc)
Thrombocytopenia	Clinically significant; severe cases reported	Less frequent; dose-dependent, higher at 80 mg
Injection-site reactions	Very common (~24% of injections)	Less frequent (~10–16%)
Liver enzyme elevations	Not dominant signal	Increased, especially at higher dose (80 mg)
Hepatic fat	No consistent increase	Dose-dependent increase observed
Hypersensitivity / immune	Rare (e.g., serum sickness)	Mild hypersensitivity events slightly increased
AE rate	Higher discontinuation vs placebo	Similar to placebo overall
Key safety driver	Systemic exposure / high dose	Residual dose-dependent toxicity, reduced by targeting

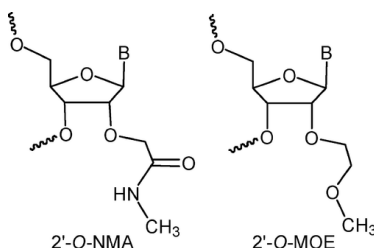
IONIS-HTTRx (Tominersen, Roche/Ionis)



In the Phase III GENERATION HD1 trial, patients received Tominersen via intrathecal injection at doses of 120 mg every 8 weeks or every 16 weeks. Although the drug successfully lowered huntingtin protein levels, it failed to improve clinical outcomes and in some cases worsened disease progression, particularly with more frequent dosing. This lack of efficacy is thought to result from non-selective reduction of both mutant and normal huntingtin along with possible overdosing and intervention occurring too late in the disease course.

Nusinersen (Spinraza®)

Nusinersen (Spinraza®) is a fully modified 2'-O-methoxyethyl (MOE) antisense oligonucleotide with a phosphorothioate backbone, designed to modulate SMN2 splicing by promoting inclusion of exon 7. In the ENDEAR (NCT02193074) and CHERISH (NCT02292537) clinical trials, this sterically blocking MOE chemistry demonstrated substantial therapeutic benefit in patients with spinal muscular atrophy (SMA), leading to significant improvements in motor function and survival compared with untreated cohorts (Finkel et al., 2017; Mercuri et al., 2018). The clinical efficacy of nusinersen reflects key properties of fully modified MOE ASOs, including high binding affinity, strong resistance to nuclease degradation and prolonged tissue residence within the central nervous system following intrathecal administration (Geary et al., 2015). This durability supported maintenance dosing at four-month intervals and established MOE-modified splice-switching oligonucleotides as a clinically validated platform for CNS splicing modulation.



However, because CNS-directed SSOs must be administered intrathecally, further improving potency and durability remains an important therapeutic goal. Longer-lasting molecules could reduce the frequency of

lumbar punctures, lessen treatment burden for patients and caregivers and potentially improve clinical outcomes by sustaining target engagement over extended periods. These considerations have driven the development of next-generation ribose modifications that retain the favourable properties of MOE while improving pharmacological efficiency. One such chemistry is 2'-O-[2-(methylamino)-2-oxoethyl] (NMA), a 2'-O-substituted sugar modification structurally related to MOE but distinguished by an amide-containing side chain. NMA originated from efforts to improve second-generation ASO chemistries by enhancing RNA binding, nuclease resistance and metabolic stability while maintaining tolerability. Early studies showed that NMA-modified oligonucleotides could be efficiently synthesized and incorporated into ASOs, preserving strong hybridization to complementary RNA and showing pharmacological activity comparable to MOE-modified analogues (Prakash et al., 2003; Pattanayek et al., 2004; Prakash et al., 2008).

More recent work has extended NMA chemistry to splice-switching oligonucleotides, where it appears to offer advantages over MOE in CNS-relevant applications. In preclinical models of SMA, NMA-modified SSOs targeting the same SMN2 splicing regulatory region as nusinersen produced greater exon 7 inclusion at lower doses, with optimized NMA candidates showing approximately three- to four-fold higher potency than MOE comparators while maintaining prolonged activity in CNS tissues (Ling et al., 2026). This improved activity was not explained by increased tissue accumulation, suggesting that NMA enhances intracellular pharmacology rather than simply improving delivery or exposure. Potential mechanisms include altered intracellular trafficking, improved nuclear availability or modified interactions with RNA–protein complexes involved in splicing (Ling et al., 2026). Collectively, these findings position NMA as a next-generation chemistry for steric-blocking ASOs, with the potential to improve potency, extend duration of action and reduce dosing frequency for intrathecally administered CNS splicing therapies.